Math Check #5 — Write Ms. Womack's Model

OPIM 5641 · Studio 5 · Nov 4, 2026
5 points · from memory, on paper

Name: _____

Write the model in math, using x, $\bar{r}$, Cov, and y. Words alone do not count.

Part 1 · Imagine you have two stocks, A and B. How much money goes into each?

1. Decision variables what are they, and what does $x_A = 0.25$ mean?

2. Objective function maximize expected return

3. Constraint: invest all the money

4. Constraint: keep risk under the limit write all 4 terms of the weighted covariance matrix

5. Constraint: each proportion is between 0 and 1

Part 2 · Now add business rules, with many stocks: A, B, C, and so on. Write each rule for stock A.

6. The new binary decision variable for stock A what does it equal, and when?

7. Linking constraint: if we buy stock A, at least 2%

8. Linking constraint: never more than 10% in stock A

9. A, B, and C are our three Tech stocks. Pick at least 2 of them.

10. With the 10% cap, what is the fewest stocks you can hold, and why?